Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/06/2005
PI Name:	Lupton, Joanne R. Ph.D.		
Project Title:	Gene Expression Patterns Predictive of Radiation-Enhanced Colon Tumorigenesis: Diet as a Countermeasure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-2253	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Turner, Nancy D. (Texas A&M University) Chapkin, Robert S (Texas A&M University) Braby, Leslie A (Texas A&M University) Ford, John R (Texas A&M University) Carroll, Raymond J (Texas A&M University)		
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Task Description:	There is currently no reliable, non-invasive method to determine the response to radiation exposure and development of colon cancer throughout the disease process. One potential method to monitor the changes in colonocyte gene expression that occur during the various stages of the disease would be to use mRNA from sloughed cells in the feces. This proof of principle project has as its main goal the modification of existing protocols in order to improve the effectiveness of polyA isolation from feces in preparation for analysis of gene expression using microarray techniques. Samples for this project were collected from rats treated with 1 Gy, 1 GeV/nucleon Fe ions at Brookhaven National Laboratory, followed by injection with a colon carcinogen (azoxymethane, AOM) 10 and 17 days after radiation exposure or from rats treated only with AOM. The rats consumed diets that have been demonstrated to either be promotive or protective against methylation- and oxidation-induced colon cancer. Fecal samples collected at 7, 14, and 28 weeks after the last AOM injection were used to determine changes in gene expression using microarray technology. The data collected from this project will determine whether improvements to existing protocols make it feasible to use isolated fecal polyA for to measure gene expression over time in order to document radiation exposure and colon cancer development in a non-invasive manner. The specific aims of this project are to: #1. Optimize polyA isolation and sample handling steps prior to array hybridization in order to develop a protocol that could reliably be used to measure gene expression from exfoliated colonocytes. #2. Develop a normalization procedure that would be robust enough to deal with contaminated or degraded RNA samples. Routine screening throughout long-term missions will be critical if the monitoring of radiation exposure and carcinogenesis induction is to be successful. A non-invasive procedure would make sampling easier and less stressful on the astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The gene expression analyses should contribute towards the development of a non-invasive screen for colon cancer that can be applied to everyone.
Task Progress:	Introduction: This proof of principle project has as its main goal the modification of existing protocols in order to improve the effectiveness of polyA isolation from feces in preparation for analysis of gene expression using microarray techniques. Specific Aims. #1. The first aim was to optimize polyA isolation and sample handling steps prior to array hybridization in order to develop a protocol that could reliably be used to measure gene expression from exfoliated colonocytes. #2. The second aim involved development of a normalization procedure that would be robust enough to deal with contaminated or degraded RNA samples. The CPR question most applicable to this research is 31d “How can effective biomarkers of carcinogenic risk from space radiation be developed and validated?” The work conducted addresses this question through the development of protocols that would enable biomarker identification for radiation exposure and colon carcinogenesis. Routine screening throughout long-term missions will be critical if the monitoring of radiation exposure and carcinogenesis induction is to be successful. A non-invasive procedure would make sampling easier and less stressful on the astronauts. Therefore, a technique using sloughed cells to determine the exposure to radiation and the response to that exposure, as measured by changes in gene expression, would permit routine screening. This research did not identify any new CPR questions. The risks to personnel in space from naturally occurring radiations are generally considered to be one of the most serious limitations to human space missions (BEIR 1990; BEIR 1998; Williams et al. 1999) and are considered a Tier One problem by NASA. Of all the risks of radiation exposure, the highest priority is development of cancer. Although definitive epidemiological data are not available on which types of cancer may be enhanced by radiation exposure (due to low numbers of astronauts compared to numbers required for statistical significance) one study, in pilots, shows an added risk for colorectal cancer (Rodionov and Potapov 2002). This is not surprising since colon cancer is the second leading cause of death from cancer in the US today and strikes men and women nearly equally (Landis et al. 1998). Colon cancer can be induced by chemical carcinogens or by radiation, but the combination of the two appears to be at least additive or possibly synergistic. Fortunately, of all the cancers, colon cancer is the most amenable to dietary intervention and one of the easier to cure if detected early enough in the disease process. We had already developed a noninvasive technique to analyze changes in colonocyte gene expression from colon cells exfoliated into fecal material through standard PCR techniques. The goal of this work was to improve upon the existing protocol so that the isolated material could be used for global transcriptional profiling over time to document radiation exposure and colon cancer development in a non-invasive manner. Materials & Methods Weanling Sprague-Dawley rats were provided diets containing either fish oil or corn oil as the lipid source and either cellulose or pectin as the fiber source. Half of the rats were exposed to 1 Gy of 1 GeV/nucleon Fe ions at Brookhaven National Laboratory 3 weeks after starting the experimental diets. All rats received injections of a colon-specific carcinogen (azoxymethane, AOM) on days corresponding to 10 and 17 days after irradiation. Fecal samples were collected 7, 14, and 28 weeks after the last AOM injection. Mammalian polyA RNA was isolated from fecal pellets collected soon after defecation, stored in RNA protection buffer and frozen (-80°C). The polyA was used to generate cDNA, which was amplified through PCR. The amplified cDNA was used to generate biotin labeled cRNA via in vitro transcription techniques as described by GE Healthcare. Quality and quantity of polyA and of cRNA were assessed on an Agilent Bioanalyzer. The cRNA was hybridized to GE CodeLink Rat genome arrays using standard protocols. Results were normalized to minimize systematic variation in raw gene expression levels so that across slide comparisons could be conducted and biological differences identified. We developed a two-stage method, following robustness principles that perform a global normalization at stage one and an intensity-dependent normalization at stage two. Results: Specific Aim #1 Modifications to the polyA RNA isolation procedures improved the quality and quantity of the polyA recovered. A comparison of the polyA isolated from rat feces and a standard polyA demonstrates a fairly similar distribution in size of

material recovered. Even though there is still residual bacterial RNA found in the isolated mammalian polyA, the resulting sample is highly enriched in the mammalian material.

An analysis of the quality and distribution of the cRNA generated by the procedure demonstrate highly comparable results obtained with the standard polyA sample and the fecal sample. The net increase in the cRNA generated by the combined new procedures was approximately 100-fold.

Arrays generated using the cRNA from fecal polyA had consistent spot quality within and across arrays and there were reasonable distributions in the level of staining present. There were obvious signs of contamination in certain spots across arrays and the normalization techniques developed for specific aim #2 were developed to deal with these issues.

Of the 33,000 genes on the array, an average of 13,000 are expressed at robust levels, with 19,000 genes having low to negligible expression. It was possible to identify differentially expressed genes in response to treatments.

Specific Aim #2

Our initial hypothesis that the issues of potential sample degradation and contamination would require a new technique was justified. A novel normalization procedure has been developed and is in the process of validation and comparison with other procedures that have been reported for use in partially degraded samples. The procedure is capable of identifying spots that should be eliminated from further analyses.

Bibliography Type:	Description: (Last Updated: 08/22/2020)
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